

Architecture Description

Housing Affordability Forecasting System – Greater Darwin Region

PRT661 – Data Science Practice | Group Dan1

This document describes the high-level system architecture shown in Figure 4.1 of the Assessment 1 report. It explains each stage of the pipeline and how data flows from the original sources through to the final dashboard used by end users.

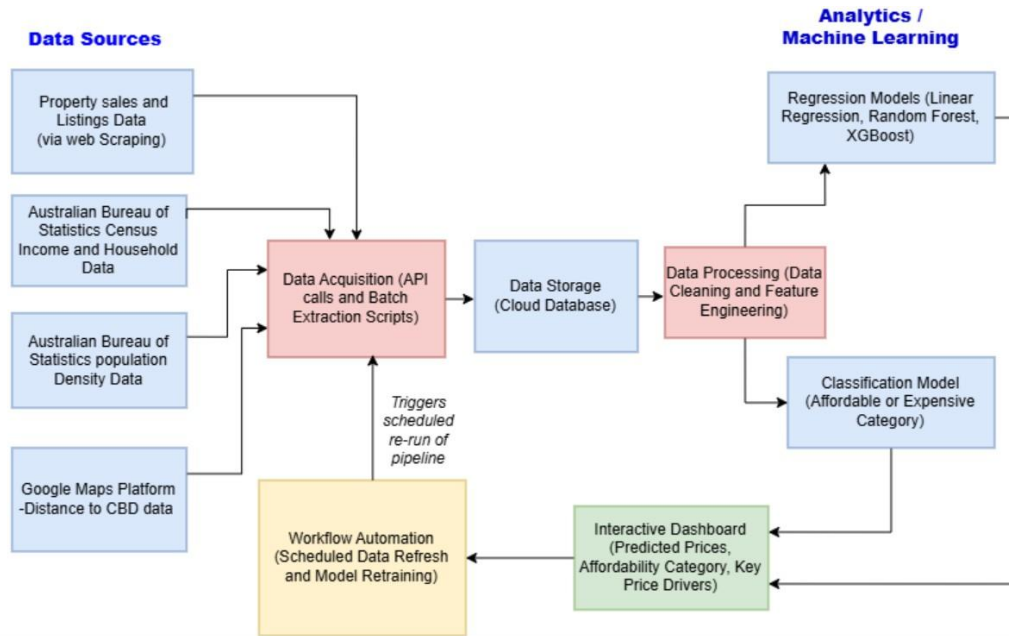


Figure 4.1: High-Level System Architecture – Housing Affordability Forecasting System

1. Data Sources

The system draws on four external data sources. Property sales and listing data is collected via web scraping, providing the core price records used for prediction. The Australian Bureau of Statistics supplies two datasets: census income and household data, and population density data, both of which provide suburb-level demographic context. Finally, the Google Maps Platform is used to calculate distance-to-CBD, a key locational feature.

2. Data Acquisition

Data is pulled from these sources through a combination of API calls and batch extraction scripts. This stage is responsible for retrieving raw records on a scheduled basis and passing them into storage without transformation, so that the original source data is always preserved.

3. Data Storage

Raw and processed data is held in a cloud database. This centralises all four source datasets in one place so that later stages of the pipeline can query a consistent, unified dataset rather than accessing each source separately.

4. Data Processing

Before modelling, the data is cleaned and transformed. This includes removing duplicates, handling missing values, standardising suburb, date and price fields, and engineering new features such as distance from CBD, price per square metre, and suburb income indicators. The output of this stage is a processed, model-ready dataset.

5. Analytics / Machine Learning

The processed dataset feeds two parallel modelling approaches. Regression models (Linear Regression, Random Forest and XGBoost) are trained to predict continuous house prices. A classification model is separately trained to categorise suburbs as “Affordable” or “Expensive” based on the predicted price, giving users a quick, interpretable summary alongside the detailed price forecasts.

6. Interactive Dashboard

Model outputs are surfaced through an interactive dashboard, allowing users to explore predicted prices, view affordability categories by suburb, and identify the key factors driving price differences across the Greater Darwin Region.

7. Workflow Automation

A scheduling component automates the refresh of source data and periodic retraining of the models, triggering the pipeline to re-run from data acquisition through to the dashboard so that predictions stay current as new property and demographic data becomes available.